

SOLUTIONS

1. C is correct. The coupon payments are 5.35 ($=10.7/2$), the discount rate is 5.8 percent ($=11.6\%/2$) per period, and the number of periods is 16 ($=8 \times 2$). Using Equation 6, the calculation is as follows:

$$95.39 = \frac{5.35}{1.058} + \frac{5.35}{1.058^2} + \frac{5.35}{1.058^3} + \frac{5.35}{1.058^4} + \dots + \frac{5.35}{1.058^{14}} + \frac{5.35}{1.058^{15}} + \frac{105.35}{1.058^{16}}$$

Alternatively, using the Microsoft Excel or Google Sheets PV function (PV(0.058, 16, 5.35, 100, 0)) also yields a result of MXN95.39.

A is incorrect. MXN95.47 is the result when incorrectly using coupon payments of 10.7, a discount rate of 11.6 percent, and 8 as the number of periods.

B is incorrect. MXN97.18 is the result when using the correct semiannual coupons and discount rate, but incorrectly using 8 as the number of periods.

2. A is correct. The present value of the mortgage is NZD3.75 million ($=0.75 \times 5,000,000$), the periodic discount rate is 0.004 ($=0.048/12$), and the number of periods is 60 ($=5 \times 12$). Using Equation 8,

$$A = \$70,424 = \frac{0.4\% (\text{NZD}3,750,000)}{1 - (1 + 0.4\%)^{-60}}$$

Alternatively, the spreadsheet PMT function may be used with the inputs stated earlier.

B is incorrect. NZD93,899 is the result if NZD5 million is incorrectly used as the present value term.

C is incorrect. NZD71,781 is the result if the calculation is made using 4.8 percent as the rate and 5 as the number of periods, then the answer is divided by 12.

3. B is correct. Mylandia's next expected dividend is CAD2.472 ($=2.40 \times 1.03$), and using Equation 14,

$$PV_t = \frac{2.40(1 + 0.03)}{0.08 - 0.03} = \frac{2.472}{0.05} = 49.44.$$

4. C is correct. Following the first step, we observe the following expected dividends for Mylandia for the next three years:

$$\text{In 1 year: } D_1 = \text{CAD}2.64 (=2.40 \times 1.10)$$

$$\text{In 2 years: } D_2 = \text{CAD}2.90 (=2.40 \times 1.10^2)$$

$$\text{In 3 years: } D_3 = \text{CAD}3.19 (=2.40 \times 1.10^3)$$

The second step involves a lower 3 percent growth rate. At the end of year four, Mylandia's dividend (D_4) is expected to be CAD3.29 ($=2.40 \times 1.10^3 \times 1.03$). At this time, Mylandia's expected terminal value at the end of three years is CAD65.80 using Equation 17, as follows:

$$E(S_{t+n}) = \frac{3.29}{0.08 - 0.03} = 65.80.$$

Third, we calculate the sum of the present values of these expected dividends using Equation 16:

$$PV_t = \frac{2.64}{1.08} + \frac{2.90}{1.08^2} + \frac{3.19}{1.08^3} + \frac{65.80}{1.08^3} = 59.71.$$

5. C is correct. The FV of the bond is CHF95.25, the PV is CHF89, and the number of annual periods (t) is 3. Using Equation 18,